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$$\begin{aligned}\frac{(1+i)^2}{(1-i)^3} + \frac{(1-i)^2}{(1+i)^3} &= \frac{1+2i-1}{(1-2i-1)(1-i)} + \frac{1-2i-1}{(1+2i-1)(1+i)} \\&= \frac{2i}{-2i(1-i)} + \frac{-2i}{2i(1+i)} = \frac{1}{-1+i} + \frac{-1}{1+i} \\&= \frac{2}{(-1-1) + (-1+1)i} = \frac{2}{-2} = -1\end{aligned}$$

References